

cases. Some of the key services it offers are listed below.

▪ **Azure cognitive services**

This is a collection of APIs and SDKs that enable developers to add cognitive capabilities, such as vision, speech, language, knowledge, and search, to their applications. For example, Azure Computer Vision API can analyse images and videos for content, emotion, face detection, object recognition, and more. Azure Speech Services can convert speech to text and text to speech, as well as translate speech across languages. Azure Language Understanding (LUIS) can extract intent and entities from natural language queries and commands. Azure QnA Maker can create conversational agents that can answer questions based on a knowledge base.

▪ **Azure Machine Learning**

This cloud-based platform enables data scientists and developers to build, train, deploy, and manage machine learning models at scale. Azure Machine Learning supports various frameworks and tools, such as PyTorch, TensorFlow, scikit-learn, and ONNX. It also provides automated machine learning, which can generate optimal models based on the data and the problem type. Azure Machine Learning Studio is a graphical interface that allows users to create and deploy machine learning models without coding.

▪ **Azure Bot Service**

This service enables developers to create and deploy intelligent chatbots that can interact with users across various channels, such as websites, mobile apps, social media, and voice assistants. Azure Bot Service supports various bot frameworks and languages, such as Bot Framework SDK, Bot Framework Composer, and Bot Framework Adaptive Dialog. It

also integrates with Azure Cognitive Services, such as LUIS and QnA Maker, to provide natural language understanding and knowledge base capabilities for the chatbots.

▪ **Azure Databricks**

This service provides a unified analytics platform based on Apache Spark, a popular open source framework for large-scale data processing and machine learning. Azure Databricks enables users to run interactive queries, stream analytics, and run machine learning workloads on massive datasets. It also integrates with various Azure services, such as Azure Data Lake Storage, Azure Synapse Analytics, Azure Machine Learning, and Azure DevOps, to provide a comprehensive data and AI solution.

Azure AI services can be used for various industries, such as banking and financial services, healthcare and medical services, learning and development (education), retail and e-commerce.

Azure OpenAI services

Azure OpenAI services are a set of cloud-based AI services that leverage the cutting-edge research and technology of OpenAI, the leading AI research organisation. These services enable developers and data scientists to build and deploy intelligent applications

that can understand natural language, generate realistic text, analyse images and videos, and more (see Table 1).

Azure OpenAI services are integrated with the Azure platform, which provides a comprehensive and secure cloud environment for building, deploying, and managing AI solutions. You can use Azure cognitive services to complement Azure OpenAI services with additional functionality and features.

By using Azure OpenAI services in the Azure platform, you can enjoy the following benefits.

Access to state-of-the-art AI: You can leverage the latest advancements in AI research and technology from OpenAI, which is constantly pushing the boundaries of natural language and computer vision.

Easy and flexible integration: You can easily and flexibly integrate Azure OpenAI services with your existing applications and workflows, using various Azure services and tools that suit your needs and preferences.

High performance and scalability: High performance and scalability can be achieved for your AI solutions using the powerful and reliable cloud infrastructure of Azure, which can handle large volumes of data and requests.

Security and compliance: You can ensure the security and compliance of your AI solutions using the robust

Table 1: The usage of various Azure OpenAI services

Azure OpenAI service	Usage
Azure OpenAI Codex	Can convert natural language into code, queries, formulas, and other executable commands for various programming languages and platforms.
Azure OpenAI GPT-3	Can generate natural language text for a wide range of tasks and domains, such as summarisation, translation, chat, and content creation.
Azure OpenAI Vision	Can perform image and video analysis, such as object detection, face recognition, scene understanding, and captioning.
Azure OpenAI Conversational AI	Can create engaging and natural conversational agents, such as chatbots, voice assistants, and social media bots.